

different portfolios. These problems have caused mean-variance optimization to be widely derided. The lack of robustness of “optimized” mean-variance portfolios is certainly problematic, but it should not take away from the main message of mean-variance investing that diversified portfolios are better than individual assets. How to find an optimal portfolio mean-variance portfolio, however, is an important question given these difficulties.

4.1. SENSITIVITY TO INPUTS

Figure 3.13 showcases this problem. It plots the original mean-variance frontier estimated from January 1970 to December 2011 in the solid line. The mean of U.S. equity returns in this sample is 10.3%. Suppose we change the mean to 13.0%. This choice is well within two standard error bounds of the data estimate of the U.S. mean. The new mean-variance frontier is drawn in the dashed line. There is a large difference between the two.

The mean-variance frontier portfolios corresponding to a target return of 12% in the original case (U.S. mean is 10.3%) and the new case (U.S. mean is 13.0%) are:

	U.S. mean = 10.3%	U.S. mean = 13.0%
U.S.	−0.0946	0.4101
JP	0.2122	0.3941
U.K.	0.4768	0.0505
GR	0.1800	0.1956
FR	0.2257	−0.0502

Previously, we didn’t have negative weights in the United States because we worked with the (constrained) optimal portfolio for a risk aversion of $\gamma = 3$. This corresponds to a target return of 11.0%. The portfolio on the frontier corresponding to a target return of 12% involves a short U.S. position of −9%. This small change in the target return and the resulting large change in the portfolio weights itself showcases the lack of robustness of mean-variance optimization.

Changing the U.S. mean to 13.0% has caused the U.S. position to change from −9% to 41%, the U.K. position to move from 48% to approximately 5%, and the French position to shrink from 23% to −5%. These are very large changes resulting from a small change in the U.S. mean. No wonder Michaud (1989) calls mean-variance portfolios “error maximizing portfolios.”

4.2. WHAT TO DO?

Change Utility

My first recommendation is not to use mean-variance utility. Investors are fearful of other risks, they care about relative performance, like catching-up-with-the-Joneses or habit utility, and they dread losses much more than they cherish